

class-1

04-01-2026 (class+Lab)

Software Engineering: a process of develop/ maintain a software

Software: set of data, instructions and that perform specific task

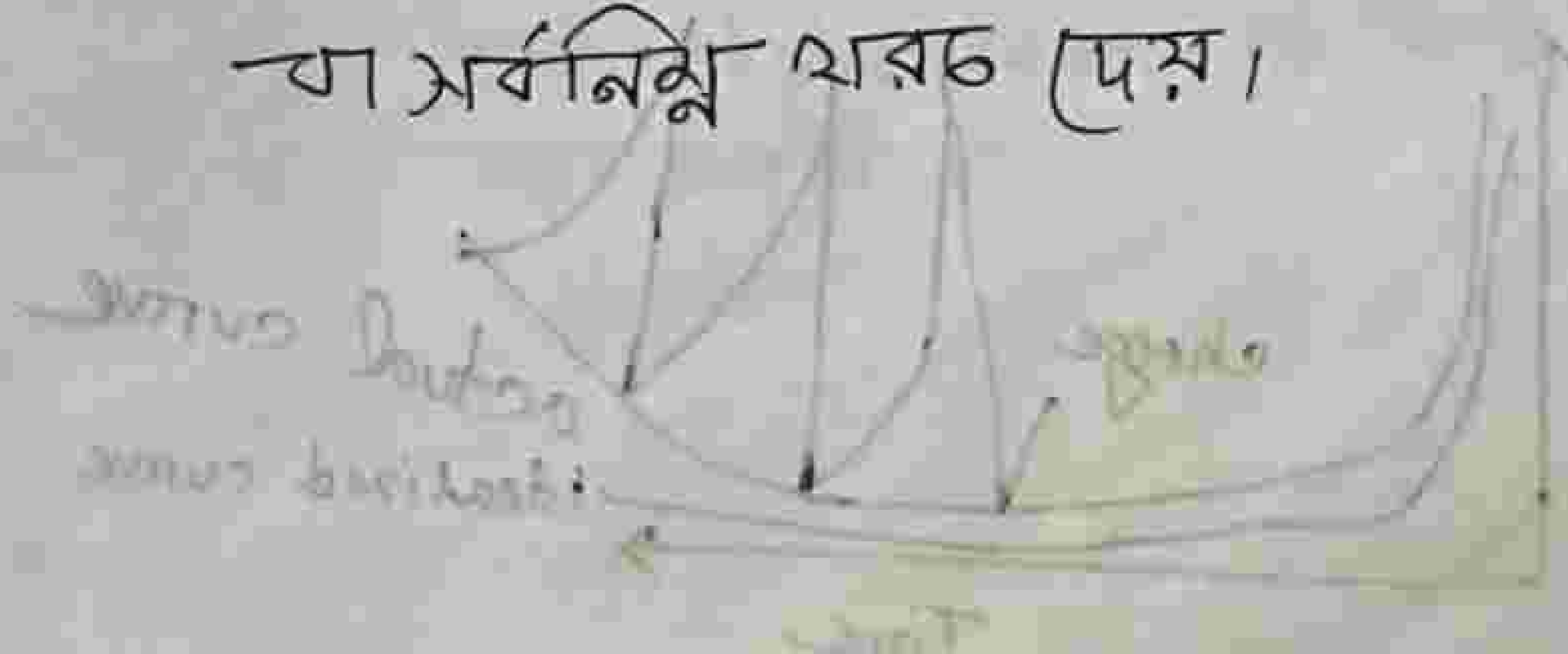
feasible & optimal → next class & ask কর
হবে।

Feasible: a solution that is practical and achievable within given constraints (time, budget, technology, resources).

যে সমস্যাটি সব constraint (কাজ) পূরণ করে।

Optimal: The best solution among all feasible ones, providing maximum efficiency, quality, or minimum cost/time.

সব feasible সমস্যার মধ্যে যেটি সর্বোচ্চ লাভ বা সর্বনিম্ন খরচ দেয়।



11-01-2026 - 10

Chapter-1:

Software Engineering: A Practitioner's

(7/e & 8/e মিনিয় পড়ণ)

*** Qstn: Duality role of a software:

1. Service as a product: software itself is delivered to users as a product. Ex- Mobile Apps, OS, web applications

2. " " " process / ~~vehicle~~ vehicle
used to create, control or support other products or systems. Ex- Compilers, Embedded control system
→ কোন software ক্রয় করা হবে নতুন
আবশ্যকীয় software বানাতে এমন। নতুন
কোন software বানান service দিচ্ছে

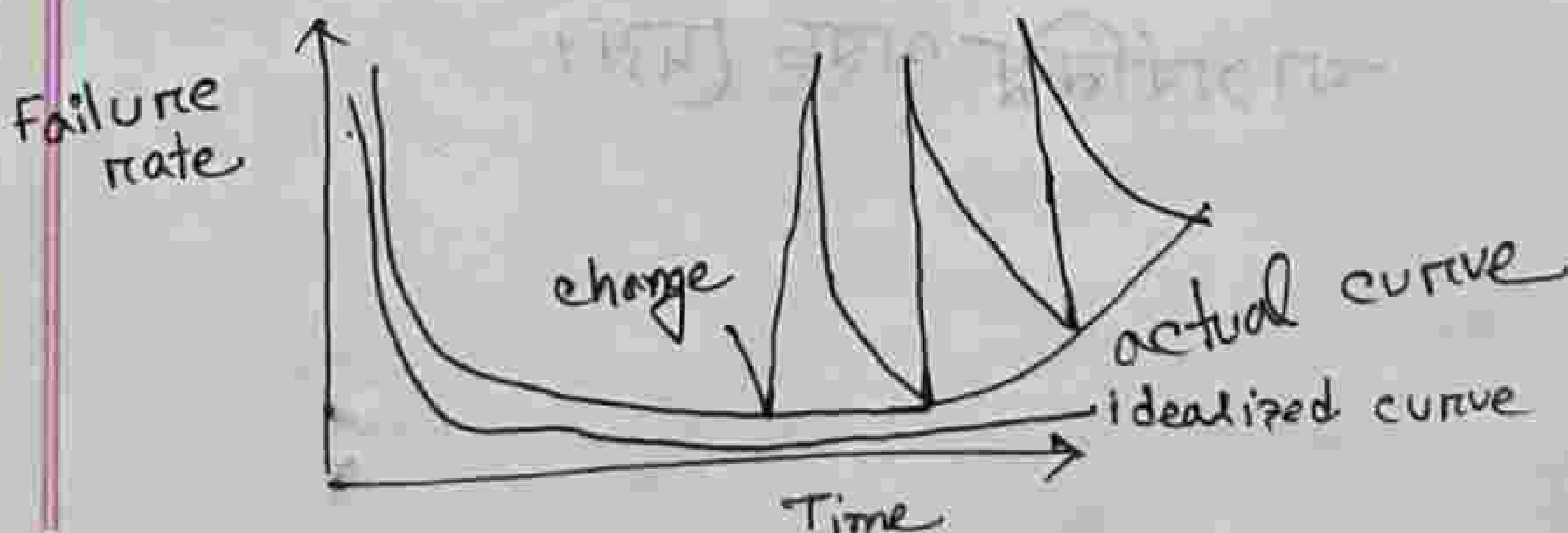
What is software?

explanation →

→ Software does not "wear out" → explain

→ সার্কিট অন্যথায় আছে সেটা নিয়ে আবার কাজ করা,

* Wear vs. Deterioration:



Qstn:

কি কি কারণে একটি certain time এর failure rate বৃদ্ধি পেতে পারে?

↳ curve খেঁচ

* Software Applications:

→ System Software

→ Application software (MS word)

→ engineering / scientific software

↳ বড় কোন mathematical term handle করতে পারে।

→ Embedded software

↳ বাহির থেকে input নেয় & user কে service দেয়। (microwave)

→ product-line software

↳ common থাকে feature কিছু

→ WebApps

↳ website, internet এর মাধ্যমে connect থাকে এমন মোকাম কিছু

→ AI software

↳ knowledge based এ কাজ করে। trained করা থাকে।

* Software - New Categories:

↳ challenges also

→ Open world computing: Cloud computing,
বিভিন্ন স্থান থেকে resource একটি জায়গায় service দেয়।

** Ubiquitous computing:

wireless network এর মাধ্যমে

→ Net sourcing: Google এ বিভিন্ন web এ তথ্য
ছড়িয়ে ছিটিয়ে আছে। Google

→ Open source: সবাই নিজের মতো edit করতে
পারে এমন। কিছু এটা হল অন্য কিছু অনৈতিক
কাজে ব্যবহৃত হতে পারে যা সমস্যা।

* Legacy Software: (Legacy challenge)

Why must it change?

Situation এর
আর feature update

→ এক সময়ের সাথে চাহিদা অনুযায়ী
update করতে হবে।

→ নতুন feature add করতে পারবে

→ modern tech. এর সাথে সাথে এবং extend করতে
পারে

↳ নতুন feature add এ cost
বেড়ে যেতে পারে যা একটি
challenge।
service দিতে হলে user নতুন ভাবে
সমস্যা সমাধান করতে পারে।

→

* Characteristics of Web Apps-I:

- Network intensiveness - ইনিষ্টি অমরক থাকা মাগা client এর মাথ।
- Concurrency - একই সময়ে multiple user ব্যবহার করতে পারবে।
- Unpredictable load : একই সাথে কতজন ব্যবহার করতে পারবে।
- Performance - একই time এ multiple users use - কমান ও performance টি থাকা যেন।
- Availability - সব সময় ব্যবহার করতে পারা কিন্তু কিছু কিছু situation এ সম্ভব নয়।
- Data driven -
- Content sensitive - related contents সুনো show করে।
- Aesthetics - প্রমা মাগা, নান্দনিক বিষয় থাকা web এ।

12-01-2026

Software Engineering

* Quality Focus:

→ Degree of goodness

→ Correctness

→ Maintainability (আমার বানানো software এ
নতুন কিছু add করতে পারা)

→ Usability (end user সহজে ব্যবহার করতে পারে
সেটা মাথায় রাখতে হবে)

→ Bedrock for any Product (জিওরোডে আসলে
কিছু বা অজুত হয়)

* Process: (Requirements)

→ What to do (কি কি করতে হবে এই system বা বিষয়)

→ Deals with activities, actions & tasks (এ

→ How type question

* Methods: (design & implementation) **

→ How to implement

→ Communication

coupling & cohesion
difference with
example

Tools:

→ Environment

→

Why software engineering is called a Layered technology?

Why we study software engineering as a CSE student & why its important?

* Software Myths: → বিজ্ঞান প্রান্ত বিবরণ

→ Management Level Myths:

→ Customer Level Myths:

প্রান্ত দ্বিতীয় customers এর।

→ Practitioner Level Myths:

Developer side প্রান্ত বিবরণ

Tools:

→ Environment

→

Why software engineering is called a Layered technology?

Why we study software engineering as a CSE student & why its important?

* Software Myths: → বিজ্ঞান বা দ্রষ্টব্য বিষয় ,

→ Management Level Myths:

→ Customer Level Myths:

দ্রষ্টব্য চিন্তা customers এর ।

→ Practitioner Level Myths:

Developer side দ্রষ্টব্য বিষয়

* A Process Framework:

→ Framework activities

→ Umbrella Activities: ৬ টি ~~সব~~ point মিলে
~~সব~~ SDLC ~~সম্প্রদায়~~ protection
দেয়।

* Framework Activities:

৬ টি point

* Umbrella activities:

লক্ষ্যে মিকস্‌ন মাঝে একত্রে
point এ আশ্রয় থেকে সন কাপড় ঢাকতে হবে
তত্ত্বিক protection দেয়। (মুদ্রক ~~কি~~ ~~আবহি~~ ৬ টি
point মিলে SDLC-সম্প্রদায়কে protect করে।

* Adapting a Process Model:

* The Essence of Practice:

→ Polya suggests:
↳ mathematician

যে সমস্যা। অর্থাৎ সমস্যা কল্পনা (মই সমস্যা)
প্রমাণে বাস্তব হবে।

Why "umbrella"
word used in
software
engineering?

* Understand the Problem:

* Plan the solution:

* Carry Out the Plan:

conform → માનનામરે

provably → પ્રમાણિત વિના

* Hooker's General Principles 96:

~~Wan~~
~~4 # 100~~
toure

quote
↓
কোড

15-01-2062026

What is system?

A system is a set of interrelated components that work together to achieve a common goal.

কোন নাম দিয়ে বলবে এটা কি system? → exam

	Week1	Week2	Week3
Planing			
Design			
Maintain			

→ used for scheduling

chapter-2:

A Generic Process Model

software process: (not as a process)

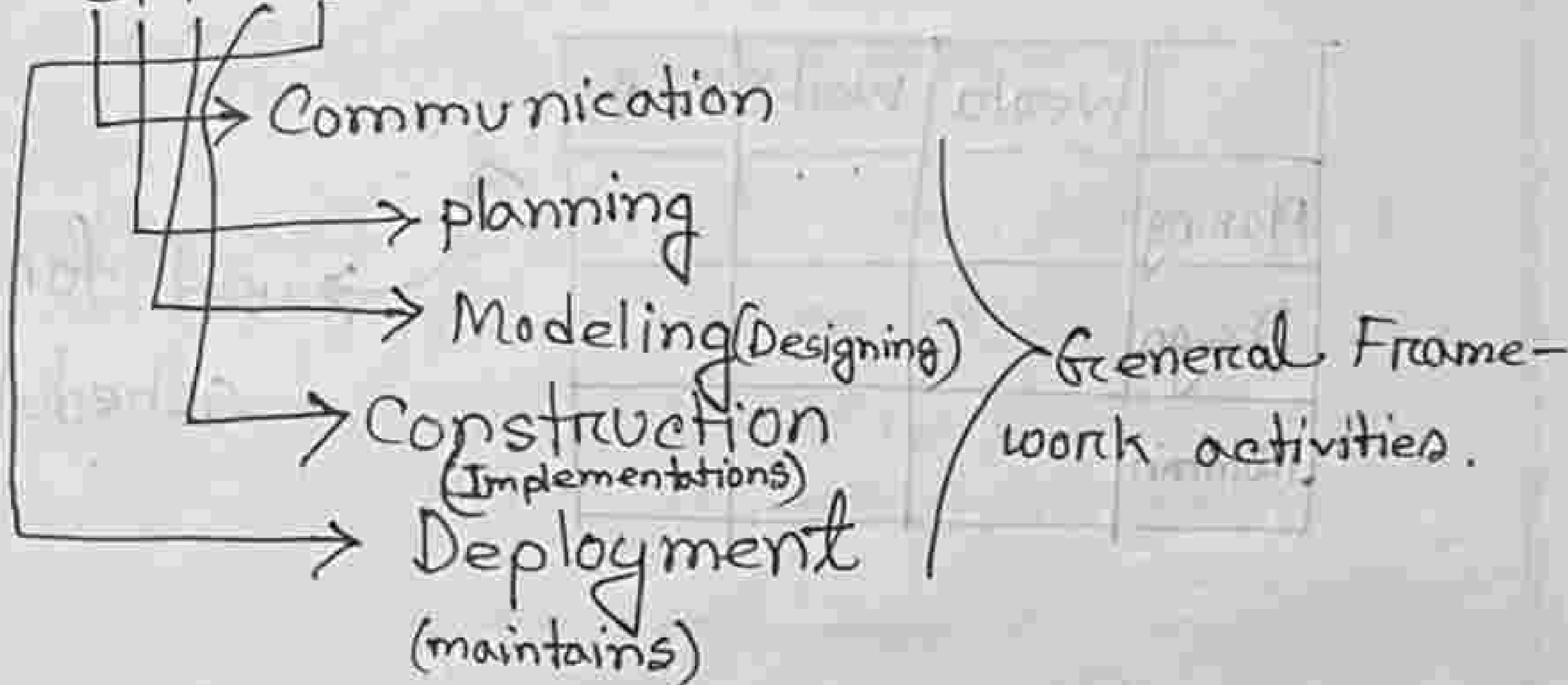
is a collection of activities, actions, and tasks that are needed to perform in order to create a software product or work product, or product.

Software process
* Sof Process Framework establishes the foundation for complete software

Engineering process.

→ Framework?

* CPMD



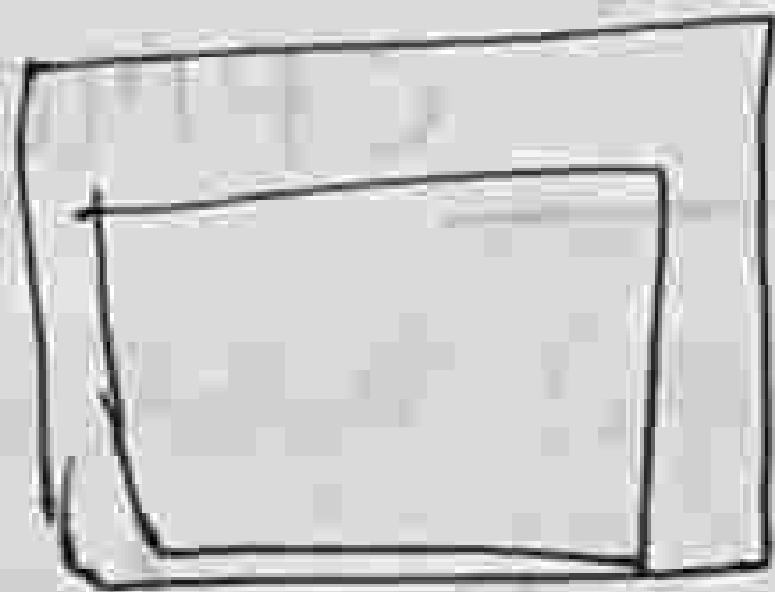
* Activity: it tries to achieve a good objective.
(a major phase of work in the software process)
ex- Testing

* Action: It encompasses a set of tasks that produce a major work product. [a small, concrete step performed to complete a task. Ex- create a test method]

* Task: It focuses on a small but well defined objective that produces a

tangible output. (a specific, assignable piece of work within an activity. Ex-write unit tests.)

*. software process figure **



Tasks set:

→ work tasks: Those are needed to be completed

→ Work ~~work~~ product: Those are going to be produced.

→ Quality ~~work~~ Assurance Points: That will be required for quality ensuring.

→ Project Milestones: Used to indicate the progress.

sunder definition
Tina 20
Exam 6

* framework activity (6 same মংস্রক action
হাকব প্রতোকব ।

* Process Flow: CPMCD

Type:

- ① Linear Process Flow (sequential)
- ② Iterative " " (" ")
- ③ Circular Process Flow (evolutionary)
- ④ Parallel Process Flow

exam 6 6th
হাকব প্রতোকব ।

যেমন:
সিদ্ধি বিলি
প্রকার general

process
ক্রিয়া

18-01-2026

(photo)

Forceps : কি কি কারণে সমস্যা হচ্ছে (খটক)।

Uses of Process Pattern :

Type: stage pattern, Task Pattern,
phase pattern.

→ known words related activity problem
→ work task related

initial context: pattern কখন use করা যাবে।

Resulting Context :

Related Pattern:

(Requirement দেখা থাকলে model select.)

বিভিন্ন প্রকারে মডেল।

page-70

#

18-1-2026 (Lab)

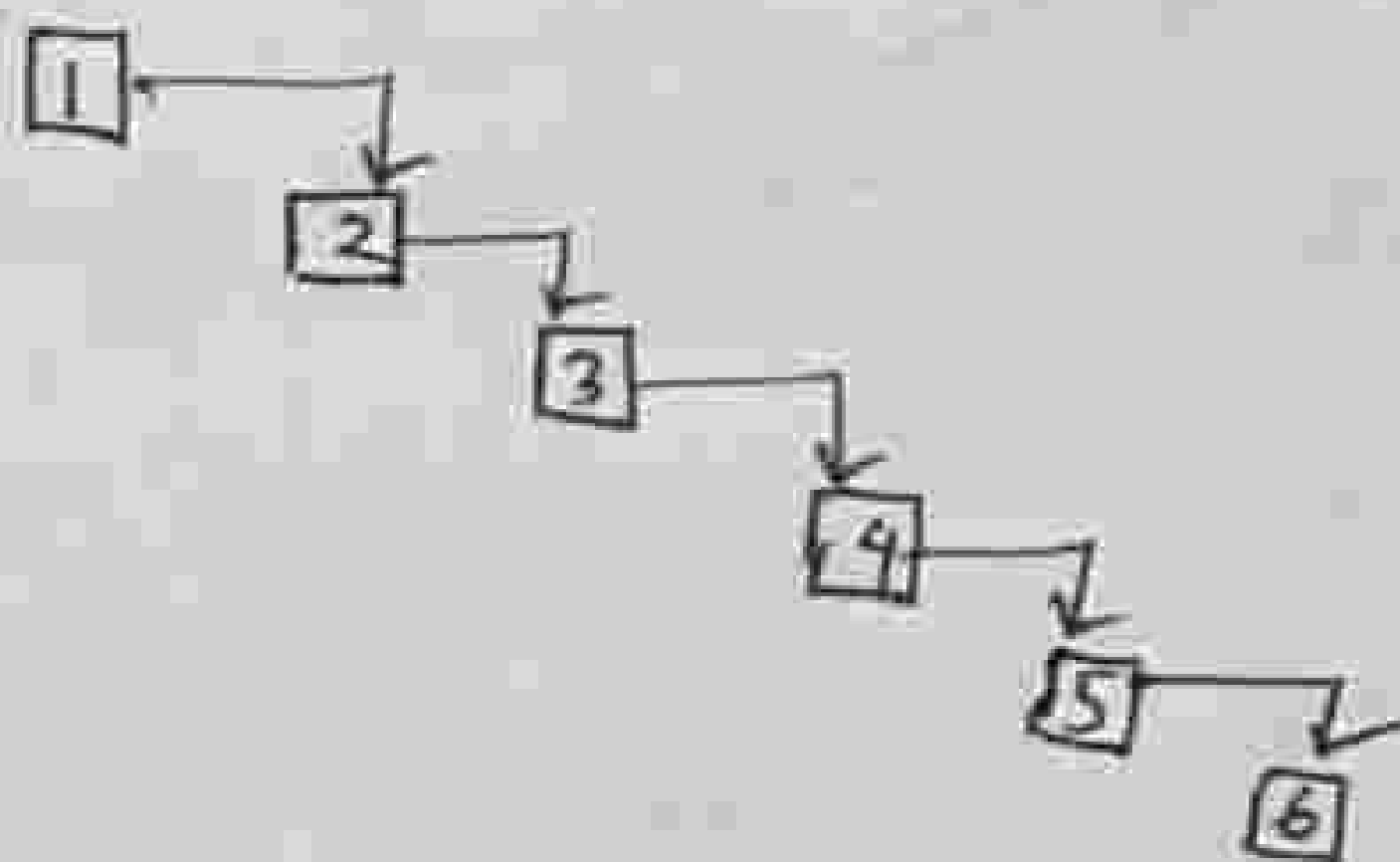
25-01-2026

prototype model: requirement clear না করা থাকলে
এতে ব্যবহার করি।

Steps of SDLC:

1. Requirement Gathering & Analysis
2. Planning
3. Design
4. Implementation & Testing
5. Deployment
6. Maintenance

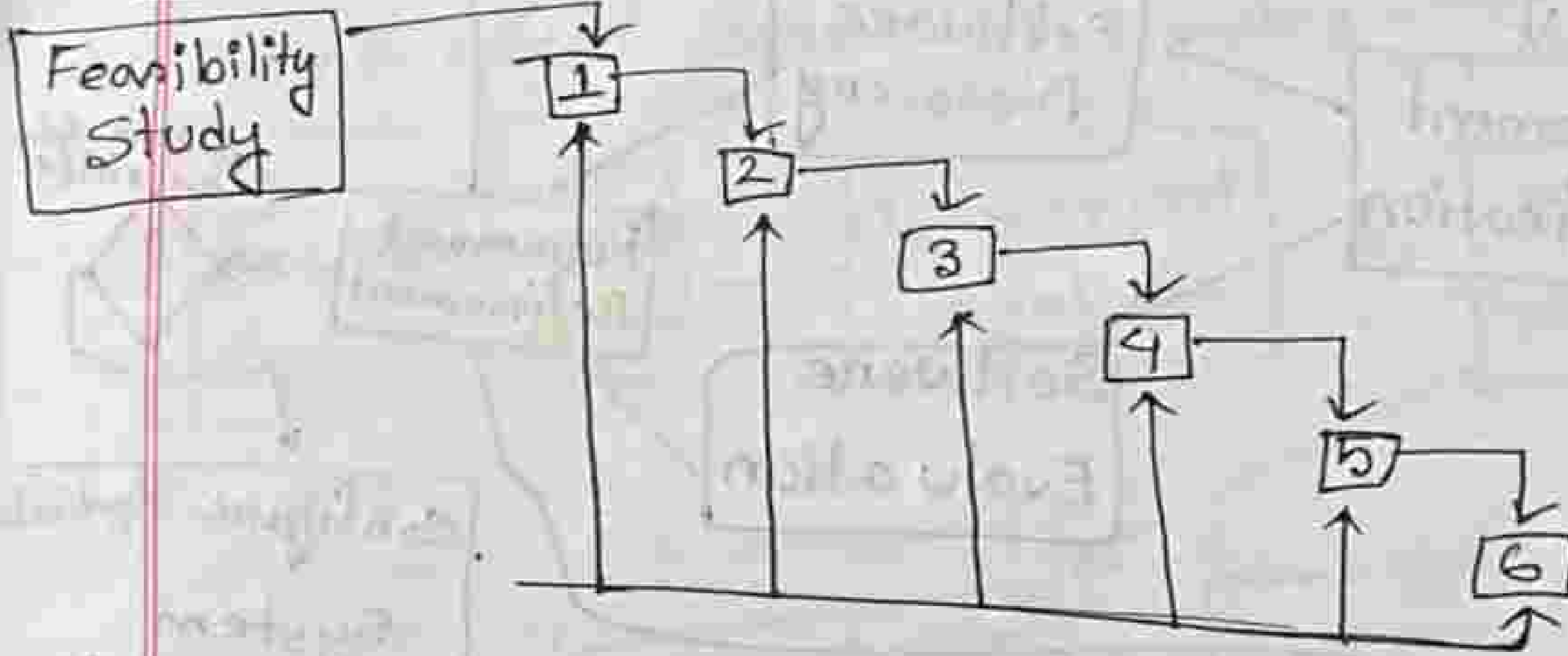
waterfall model:



Isa Mahmud

Isa Mahmud

iterative waterfall:



Incremental Model:

Incremental Model:

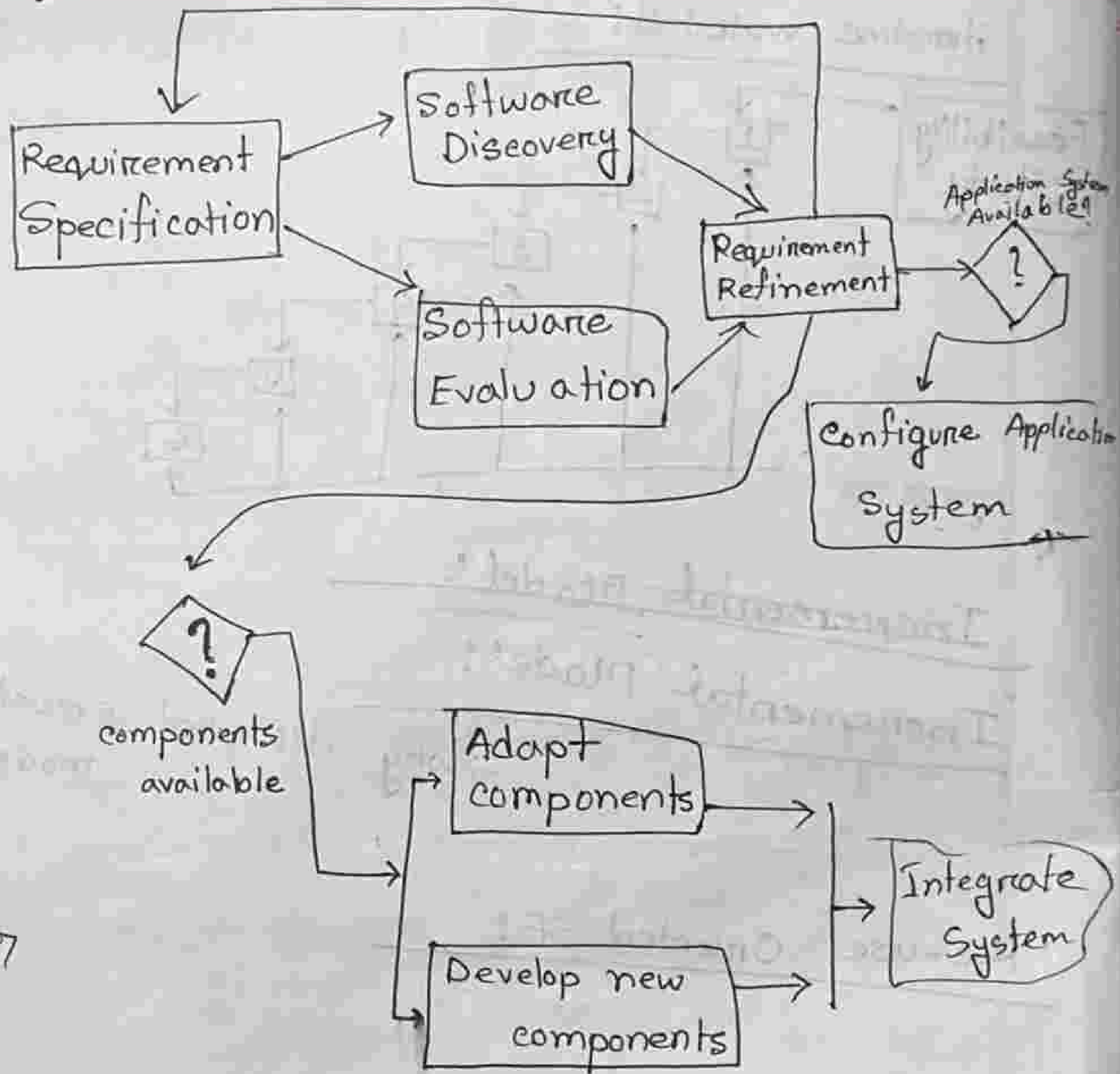
↳ why it's called evolutionary model?

Re-use Oriented SE:

Requirement
specifi

P.T.O

(CBE)
DSH



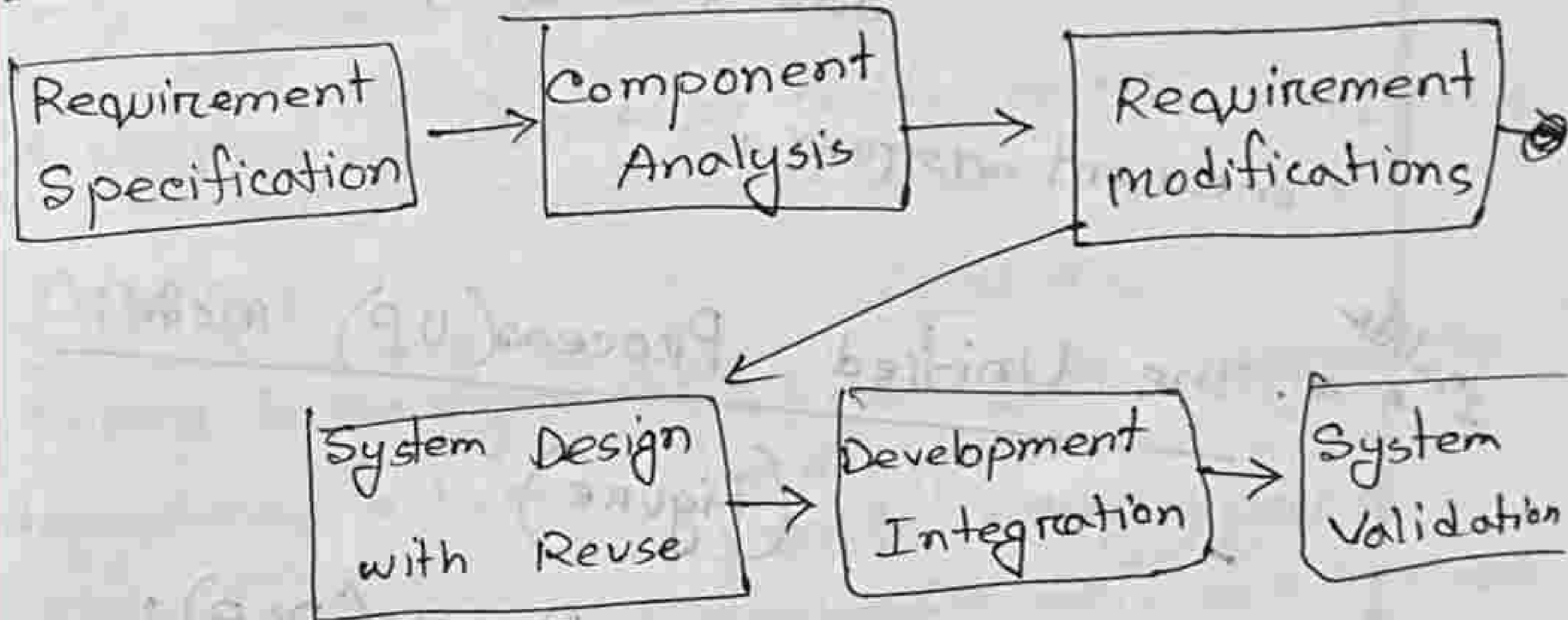
Qstn:

1. Existing components શ્રદ્ધા, ખાતર કિ કરવા?

2. " " " " " " diagram draw

29-01-2026

Component Based SE : (existing software/process & components add)



* Validation: Rules & Regulations satisfied. কখনো,
 → fulfill user needs
verification: Checking if the software is built correctly according to requirements.

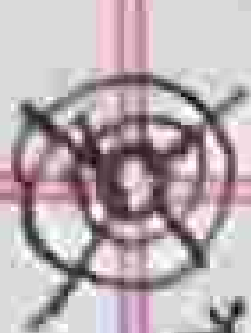
Spiral Model:



Agile Model:

Backtrack করা হবে এরকম কি কি model. প্রচুর মডি কোনগুলো efficient হবে যদি আমরা থাকে ও বড়।

continuous করে থাকবে মাঝে মাঝে software থাকবে



~~10/10/2020~~
~~10/10/2020~~

10/10/20

* Prototype ^{→ phone} iterative phase

↳ select khatam chahiye nikhaniya feature add karke shak.

Concurrent khatam.

29 slide

* The Unified Process (UP) model:

(Figure)

* Personal Software Process (PSP):

(Reading)

6 (khatam)

* Process Assessment and Improvement:

CMMI =

CMMI = Capability Maturity Model Integration

↳

↳ slide 4 khatam khatam.



New Slide = 5 pages

chapter 2 (AI)

* Coupling & Cohesion difference: (AI (2/2))

Coupling:

Coupling is the degree of dependency between modules.

- High coupling: Modules are highly dependent. (bad).
ex-Module A directly access internal data of Module B.
- Low coupling: Modules are independent (good).

ex- Module A calls a public method of module B.

Cohesion:

Cohesion is the degree to which elements inside a module belong together.

- High cohesion: module does one specific task (good).
ex- a login module only handles login related tasks.
- Low cohesion: Module does many unrelated tasks (bad).
ex- a module handles login, payment, report & email sending.

* As a CSE student why we study SE & why its important?

Soln. → To build software in a structured way.

- To handle large projects
- To improve software quality
- To reduce project Failure

→ To be industry-ready

Important: it helps a CSE student become a professional software engineer, not just a programmer

* Why umbrella word used in software engineering!

Solⁿ: Because these activities cover and support all phases of software development - just like an umbrella covers everything under it.

Umbrella activities:

→ Project Management

→ Risk management

→ Quality assurance

→ Configuration Management

→ Documentation:

traditional model: waterfall, V, incremental ~~ଅନ୍ୟାନ୍ୟ~~ ~~ଉପାଦାନ~~ ~~ପଦ୍ଧତି~~ ।

* Waterfall Model: (classic)

Feasibility study ↓

Requirement analysis
& specification ↓

Design ↓

Implementation &
Unit Testing ↓

Integration & System
Testing ↓

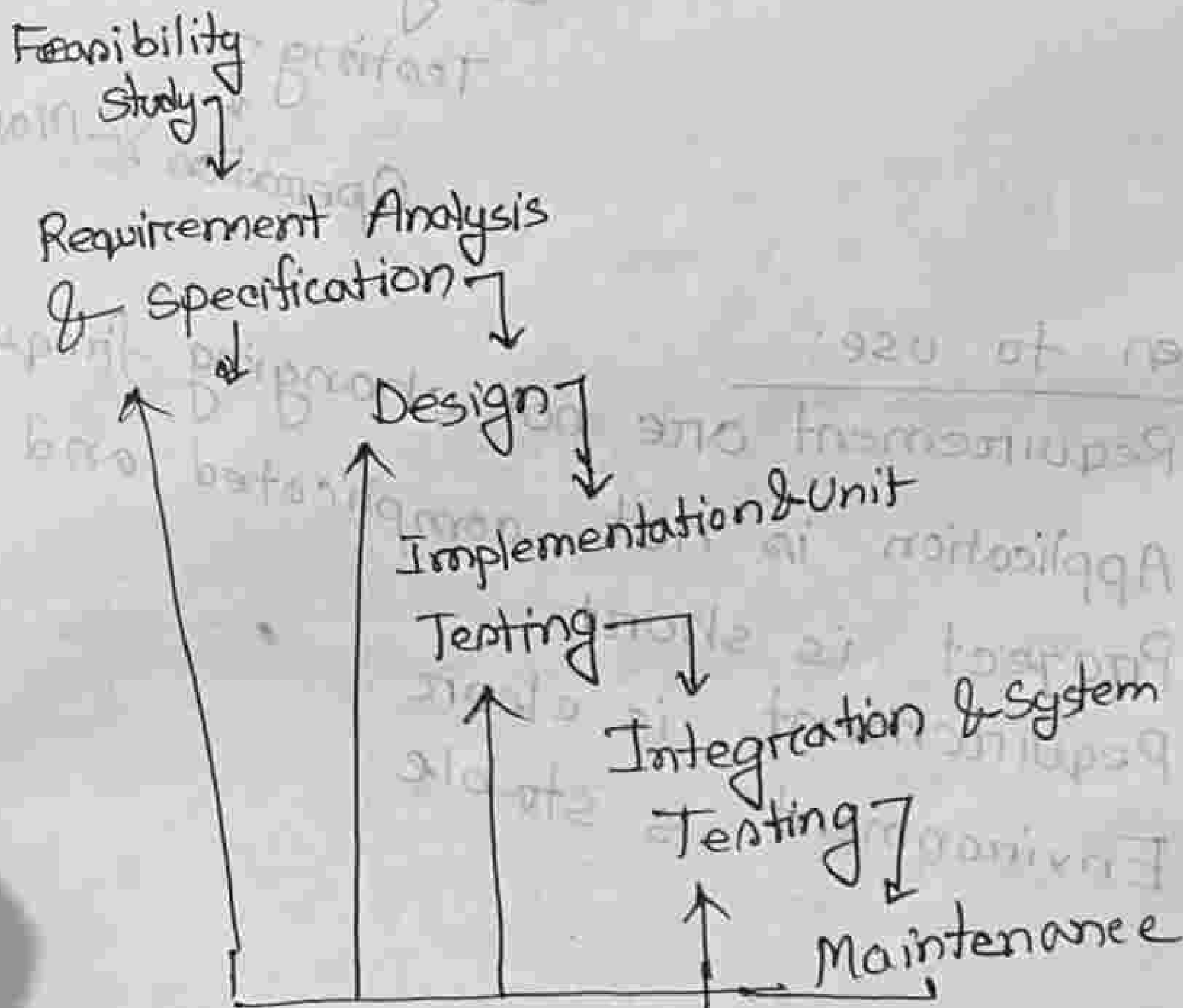
Operation & Maintenance

When to use:

- Requirement are not changing frequently.
- Application is not complicated and big.
- Project is short.
- Requirement is clear
- Environment is stable

Advantage	Disadvantage
→ Base Model → simple & easy → small projects	→ no feedback → no parallelism → no experiment → High Risk → 60% efforts Maintenance

Iterative waterfall:

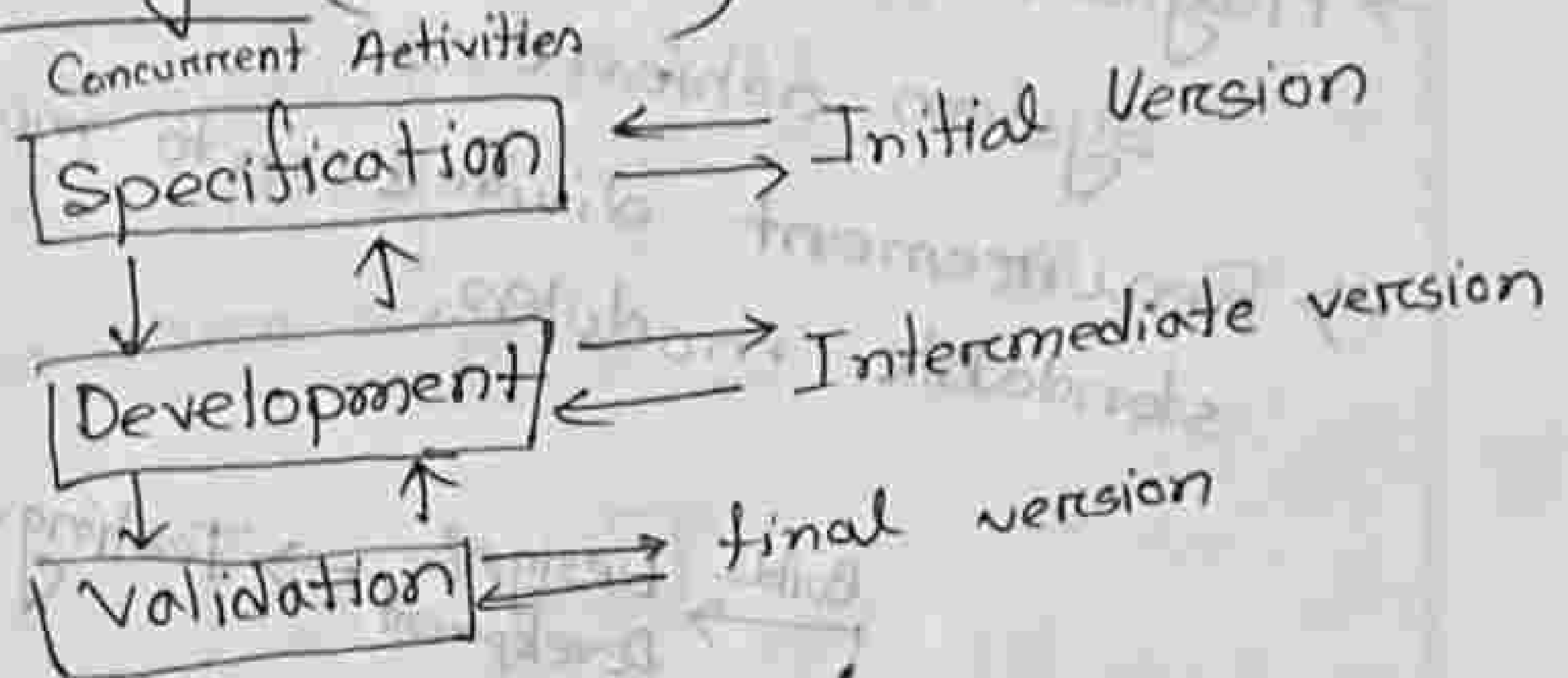


disad.

advantages same like as classic waterfall.

extra advantage = feedback

* Evolutionary: (ISD 2501)



(A)

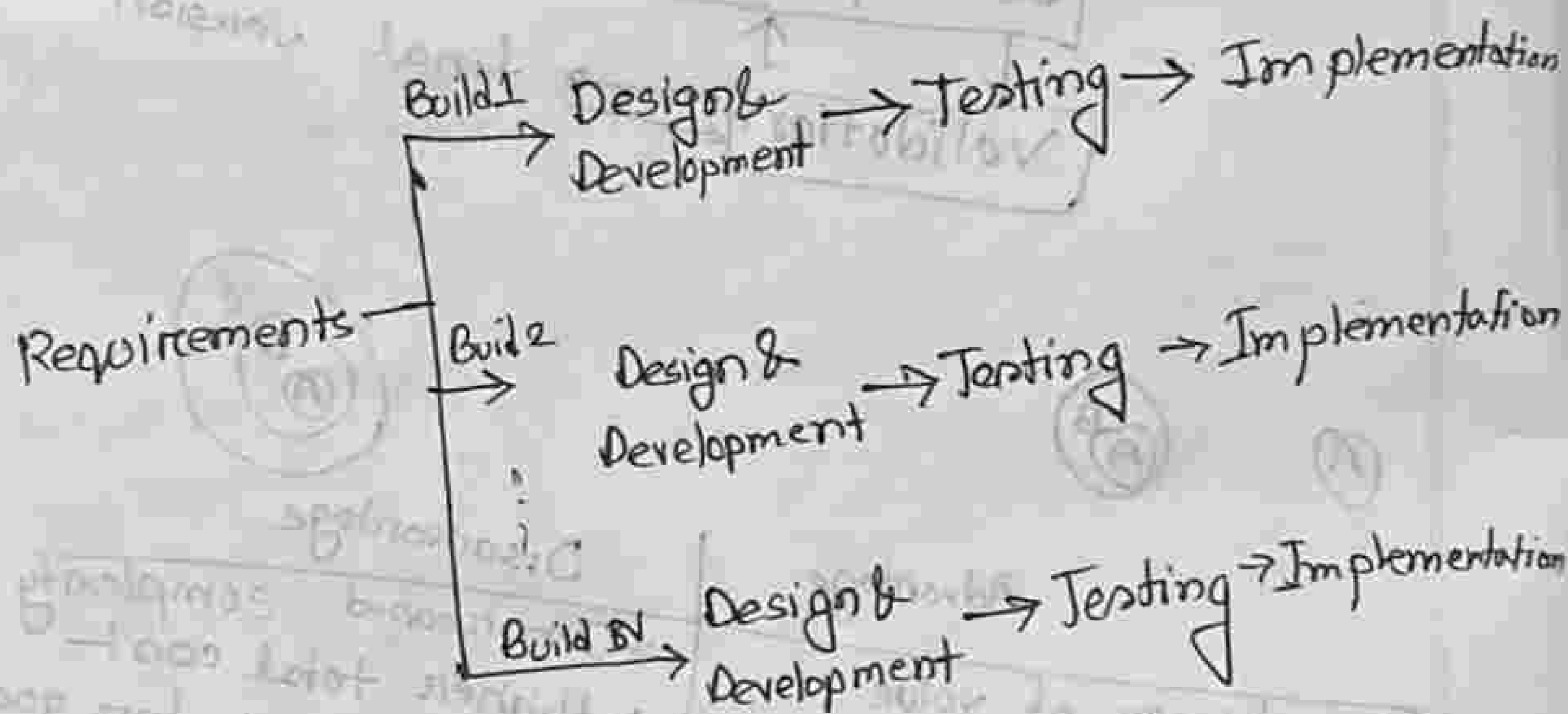
(A^B)

(A^{B^C})

Advantage	Disadvantage
→ Faster delivery of value	→ Increased complexity
→ Feasible to change	→ Higher total cost
→ Easier Risk Management	→ Requirement for good planning
→ Improved Debugging	→ Integrated challenges
→ Large projects	→ Not suitable for smaller projects
→ Customer interaction maximum	→ Highly skilled resources are required.

* Incremental Model / Development Model:

- every subsequence release of the module adds function to the previous release
- Progress continuous until the complete system achieved.
- Requirement divided into multiple standable modules.



When to use:

- Requirement are clearly classified specified, understood.
- Customer demand Quick Release.
- long development schedule.
- Software team are not very well skilled & trained.

→ web applications & Product based companies.

Advan.

- clients gets important functionality early.
- clients have a clear idea of the project
- Errors are easy to be recognized
- Easier to test & Debug
- Simple to manage risk
- changes are easy

[ISD 210]

* Advantage & Disadvantage of prototype model:

Adv.	Dad.
→ clear requirements	→ Scope creep (to many changes)
→ Early user feedback	→ It Can increase time & cost
→ Early error detection	→ Weak design if not managed well
→ Better user satisfaction	→ needs

Absent = 30 (roll)

22-02-2026

* SW Requirement Engineering (SRE/RE):

7 task

[Barry Boehm's W5HH Principle & question
সুনার answer নিয়ে কাজ।]

tasks:

→ What & How

→ user কি চাচ্ছে

→ কিভাবে চাওয়া পূরণ করতে
প্রতি system req.

* User requirement vs System Requirement.

* Function Req., Non Functional Req. &
Domain Req.

Domain Requirement: সব জায়গায় অর্থাৎ access পাওয়া

* Req. Gathering challenges.

* Need of Req. Engineering:
tools কি কি।

©

important question

RE 7 Tasks:

1. Inception: উদ্ভা ; 2. Elicitation: চাওয়া-চাওয়া
user demand বা need থেকে unnecessary ছাড়া
clear করে। বাদ দিতে।

3. Elaboration: চাওয়া-চাওয়া details থাকা আগে।

4. Negotiation: price নিয়ে deal negotiate বা feature
update আলোচনামূলক feature একতায় আগে
আগে।

5. Specification: Documentation থাকে মত

6. Validation

7. Req. Management: Rev সংশোধন-বিশোধন করে
আগে।

* SRS: → user এবং জন। তাই user বুঝে একতায়
Long ব্যবহার করবে।

⑥ Sudia, tonu, toni, ⑨ Auph, Sozid, ⑪ Nodun, ⑫ Misha, ⑬ Ehomon, ⑭ Zoruf, ⑮ Mehedi, ⑯ Mahadi, ⑰ Sabbin
 ③ 34 17

Raihan, Shalin

26-02-2026

System Process Model:

* Software System Modeling:

⇒ System perspective:

1. External perspective: System environment
कि कि निम्न ,
2. Interactive : ~~कोन एक~~ between
a system & its environment
or
between the components
of a system.
3. Structural : ↪ system &
architecture.
4. behavioral: Action, Reaction
type प्रतिक्रिया ।

↙ हर एक Perspective Represent करने के लिए
 UML Diagram Needed

Types of UML:

1. Activity Diagram: Activities involved in a data purchasing system



[किसी कार्य system या तो sub system का activity]

2. Use-case Diagram: interaction between a system & its environment.

3. Sequence Diagram: interaction between actors & the system ~~and~~ ^{and interact} between system components.

4. Class Diagram: It will show the object and classes in the system and the aggregation between the classes.

5. State Diagram: how the system reacts to internal & external events.

process or state diagram: OS steps

Ready → Running

Executed

Waiting



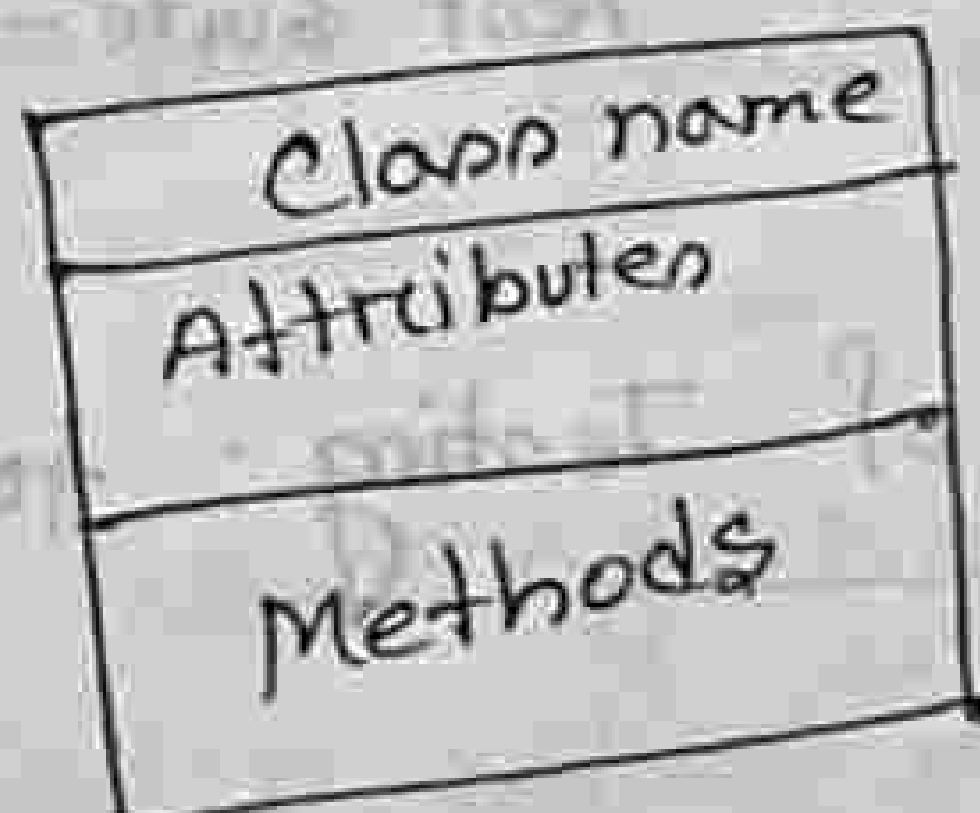
* Use-case :

বড় project এর ক্ষেত্রে module বন্য হয় যাণ free rectangle এর ভিতরে থাকে । এবং ছোট project এর জন্য process বন্য হয় । কি কি task perform করে (মর্চ) ।

* Sequence Diagram : next class

* Class Diagram :

+ → public
- → private
→ protected.



Absent: Saïmon, Meher,
Abdullah, drubho

02-03-2026 (New teacher)

09-03-2026 (online)

Software Testing:

[previous सूची देना रहे है]

→ Program Testing: expected अनुमायी behave करेगा किना (यथा)

Failure - (known or defect or bug)
- expected output यदि ना आए ,

Test suite - test case को एक साथ ,
↳ 3 step [triplet]

→ Aim of Testing: to identify all defects

important

→ Verification vs Validation:

slide 4 (पूरे को एक साथ लिखें)
in details example में
देखा गया रहे है।

→ phone के लिए बनाया किना जायगए phone checker
product बनने phone पाठित जायग। [एक- code, requirement design बना]

→ Fully deployed करेगा पर customer को
छाहिये fullfill रहे किना।

→ Design of Test Cases: check करे की test case
बनाया।

code को error found करे
design

slide 4 (यहाँ code को error
जायग वहाँ error test
case design करेगा।)

→ Design of Test Cases (cont...)

Error find করা মাঝে এমন করেই

Test suite ৩ x অবস্থান বহু তারে error find ৬ মাঝে হবে। x এর মান মাঝে
৩ তারে প্রকার করে design করা যাবে।

→ Functional Testing vs Structural Testing:

↳ blackbox testing [কোন কিছু জানিনা কিছু input দিলে মতিল output আসে]

white box → internal knowledge থাকা

imp. **

০০০

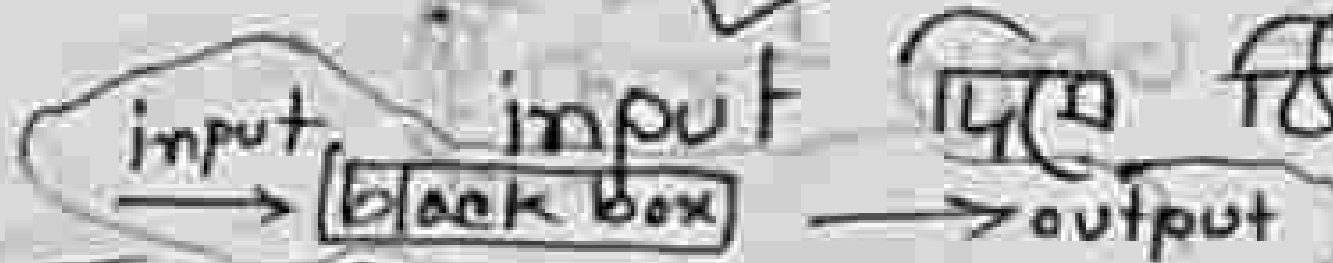
মাঝে। internal structure knowledge থাকা মাঝে।

→ Unit Testing: White box testing এর আকারে।

- প্রত্যেকটা Line/unit বহু করে test. [unit সুন্দর ওভার করা আছে।]

→ Black Box Test:

[internal knowledge থাকা নাগর না করি]



→ Equivalence Class Partitioning:

ex: একটি integer number এর square root এর
পরামা মাঝে value 0 to 5000 এর মধ্যে হতে পারে
product develop করণ।

important **

[প্রকার design ex আমাে পাও]

(math type)

[Black box vs white box]
difference & example

Test suite : (100 to 5000 এর বাহিরে, range এর মধ্যে, range এর পরামা)

→ equivalence class partitioning:

equivalence থাকা প্রকার

neg, range এর value এর বাহিরে per value test করণ।

প্রকার বহু design করে
হবে।

[White box - black box difference with example example]

ex-2:

→ Boundary Value Analysis: [] → jo karyakar programmer

⇒ দিলে দিতে পারে অঃ কাণ্ড সমস্ত handle
Tent case খুঁজা বের করতে।

0 to 5000 ରାଏ-

0.40 immediate প্রকৃষ্টিয় নহ 5000 ৳

immediate एक रात निस

ex: 0 to 5000

$$\{0, -1, 5000, 5001\}$$

→ white box testing: overall internal structural

↳ Code based Testing Test [tester ko panna → internal structure, design, implementation]

[math type astn]

code (एक line by line check कर error find.

ex₂

प्रमाण

ଆଧ୍ୟାୟ ୩

Test

ita design

code দিয়া থাকলে test case দিয়ে চ্যাক করে
 n } করে দেখবে চিকি আছে কিনা

एक प्रयोग करने के लिए एक बार

[Software Testing (एक एक भाग Question
आकर,] → final 4
→ in the mid-2

[Greeks for Greeks (एक Software Testing
all parts cover करना)]

↳ integration, Regression, Performance, Security, Alpha & Beta Test ~~अनुमान~~ just (एक)



दिना । ~~प्रश्न~~

slide ~~प्रश्न~~

Again class करे:

Software Testing strategies बार करी slide प्रस्तुत
प्रश्न (एक example सूची practice करे बना रहे) ।
प्रश्न, Google व "software engineering testing" search
दिए Greek for Greeks (एक Types of software testing
detail प्रश्न, integration, regression, performance,
security, Alpha, Beta test प्रश्न प्रश्न (एक प्रश्न दिए) ।
Advantage & disadvantage और difference .

Qstn Type:

[Equivalence partition, White Box प्रश्न (एक
Math type प्रश्न, verification & validation difference,
Black Box & white box difference] प्रश्न और [integration
Regression, performance, security, Alpha, Beta
testing] (एक example दिए प्रश्न कर type
आकरे प्रश्न) ।

प्रश्न topics (एक 1 set प्रश्न आकरे final) ।

~~Mid 2 प्रश्न प्रश्न प्रश्न~~

Mid - 2 : Software Testing part प्रश्न

30-03-2026

* Software Maintenance:

customer ^{delivery} (দেখানো) পাঠে modify করা,
update করা।

1. Corrective Maintenance:

2. Adaptive Maintenance:

যেকোন situation এ খাপ খাইয়ে
নেয়া: মতো করে।

3. Perfective Maintenance:

4. Preventive Maintenance:

* Maintenance Process Model 1:

অল্প কাজ করলে, ~~update~~

2:

*.

"

প্রত্যেকটা phase এ পরিবর্তন।

*** Estimation of approximate Maintenance
Cost:

→ ~~सिद्धि~~ न 2 नष्ट
यस लाईन नष्ट
add

যাও নারকেল delete

$$ACT = \frac{KLOC_{added} + KLOC_{deleted}}{KLOC_{total}}$$

*** maintenance cost = ACT x development cost

Exercise: $\mathbb{Q}$ is a type math 2 भाषण।

$$\text{ACT} = \frac{80 + 3.5}{55} = 1.5182$$

Maintenance cost = ACT x development cost

$$33000000 = 1.5182 \times 5000000$$

$$= 7,59,090.909$$

* Effort & Time:

organic:

semidetached: experience member

বাইরেও কিছু

Embedded: complex hardware

কোন কোন type
ওয়েব বারন
বাহ্যে
হবে।

10/12 marks
ଉତ୍ତର ଥାଏ କି ନାହିଁ
କିମ୍ବା ନାହିଁ
Formula ବୁଲାଇ
କରି ଥାଏ ।

COCOMO:

$$\text{Effort} = a_1 \times (\text{KLOC})^a$$

Organic:

$$\rightarrow \text{Effort} = 2.4 (\text{KLOC})^{1.05} \text{ PM}$$

Ex: ଉତ୍ତର ଏ ଓଗାନିକ୍ ବନା ନା ଥାଏ ନାହିଁ ବୁଲାଇ
କରି ଥାଏ type .

Maintenance:

ଆମ ଡିଭିଜନ ବନା, user କୁ satisfied
କରିବା । ଏ ବିଭାଗ ।

01-04-2026

Matrix: (Software Quality Assurance and Quality Management)

SQA:

A set of activities for ensuring quality.
একটি ongoing process যা life-cycle অর্ন্ত
চলে থাকে।

SQP: কিভাবে quality নিশ্চিত করণের plan

SQC:

SQM: কতটুকু quality ensure হবে (মানে মান)

important
গুরুত্বপূর্ণ

Elements of SQA:

→ Standards: Orga. এর কার্যকর কাজ করার
quality চিহ্ন থাকবে।

→ Education: Team members এর 6 মাস/1 বছর
অবধি training করতে হবে।

→ Vendor Management: বাইরের কিছু জিনিস
মাগে মানে manage কর।

* Verification Vs Validation:

<u>Verification</u>	<u>Validation</u>
→ Checks whether the product is built correctly → Focus on development process → Performed during development → Uses review & inspections → ensures correctness of the development → Ex- checking if the login form has username and password fields as per requirement	→ Checks whether the right product is built. → Focuses on final product. → Performed after development. → Uses testing methods → ensures that the final product fulfills user requirements → ex- checking if users can successfully login using the form.

* Equivalence Class Partitioning:

→ when lines intersect:

condition: $m_1 \neq m_2$

$$y = m_1x + c_1, \quad y = m_2x + c_2$$

[Different slopes $\rightarrow$ lines meet at exactly one point.]

ex $\rightarrow (2, 3) (-1, 4)$

→ when lines are parallel:

condition $\rightarrow m_1 = m_2 \text{ \& } c_1 \neq c_2$

[Same slope but different y-intercept $\rightarrow$ lines never meet]

ex: $(2, 3), (2, 5)$

→ when lines are coincident:

condition $\rightarrow m_1 = m_2 \text{ \& } c_1 = c_2$

[Same slope and same y-intercept $\rightarrow$ lines overlap completely
infinite intersection points]

ex: $(2, 3) (2, 3)$

Software Testing Strategies

example-1: (Using white box testing)

For the code below design 3 test suites using white box Testing (GCD program).

1. Input : $n_1 = 12, n_2 = 18 \rightarrow$ Output : $G.C.D = 6$
 [Test multiple iterations and correct divisor selection]

2. Input : $n_1 = 25, n_2 = 25 \rightarrow$ Output : $G.C.D = 25$
 [Tests prime numbers with no common divisor except 1]

3. Input : $n_1 = 13, n_2 = 13 \rightarrow$ Output : $G.C.D = 1$
 [prime values also here, where common divisor 1]

Tests equal numbers
loop runs fully.

Euclidean Algmⁿ

GCD(48, 18)

$$48 \div 18 = \text{remainder } 12$$

$$18 \div 12 = \text{" } 6$$

$$12 \div 6 = \text{" } 0$$

$$GCD = 6$$

or

G.C.D calculation

12 & 18:

12 $\rightarrow$ 1, 2, 3, 4, 6, 12

18 $\rightarrow$ 1, 2, 3, 6, 9, 18

common factors:

1, 2, 3, (6)

largest among common. So,
GCD = 6

LCM: 20, 30

$$\begin{array}{r} 2 \overline{) 20, 30} \\ 2 \overline{) 10, 15} \\ 5 \overline{) 5, 15} \\ 1, 3 \end{array}$$

$$= 2^2 \times 5 \times 3 = 60$$

গড়কে ছাড়া
দুটি সংখ্যার
সর্বোচ্চ
সাধারণ
গুণক

example-2:

What
(behavior)

How (logic/code)

Functional Testing

Structural Testing

1. Tests what the system does
2. Based on requirements/specification
3. No knowledge of code needed (Black-box testing)
4. Focuses on input $\rightarrow$ output correctness.

1. Tests how the system works internally.
2. Based on code structure
3. Requires knowledge of code (white-box testing)
4. Focuses on paths, conditions, loop in code.

~~External~~ External behavior tested
Done by testers/users

Internal stru. tested
by developers

Test Suites:

- if branch
else branch
1. Input : $n_1 = 12, n_2 = 8 \rightarrow$ Output : $LCM = 24$
[Test branches where $n_1 > n_2$]
 2. Input : $n_1 = 7, n_2 = 9 \rightarrow$ Output : $LCM = 63$
[Test branches where $n_1 < n_2$]
 3. Input : $n_1 = 10, n_2 = 10 \rightarrow$ Output : $LCM = 10$
[Test equal numbers]

[white box testing] $\rightarrow$ Test branches
[black box testing] $\rightarrow$ Test input/output

ex-5:

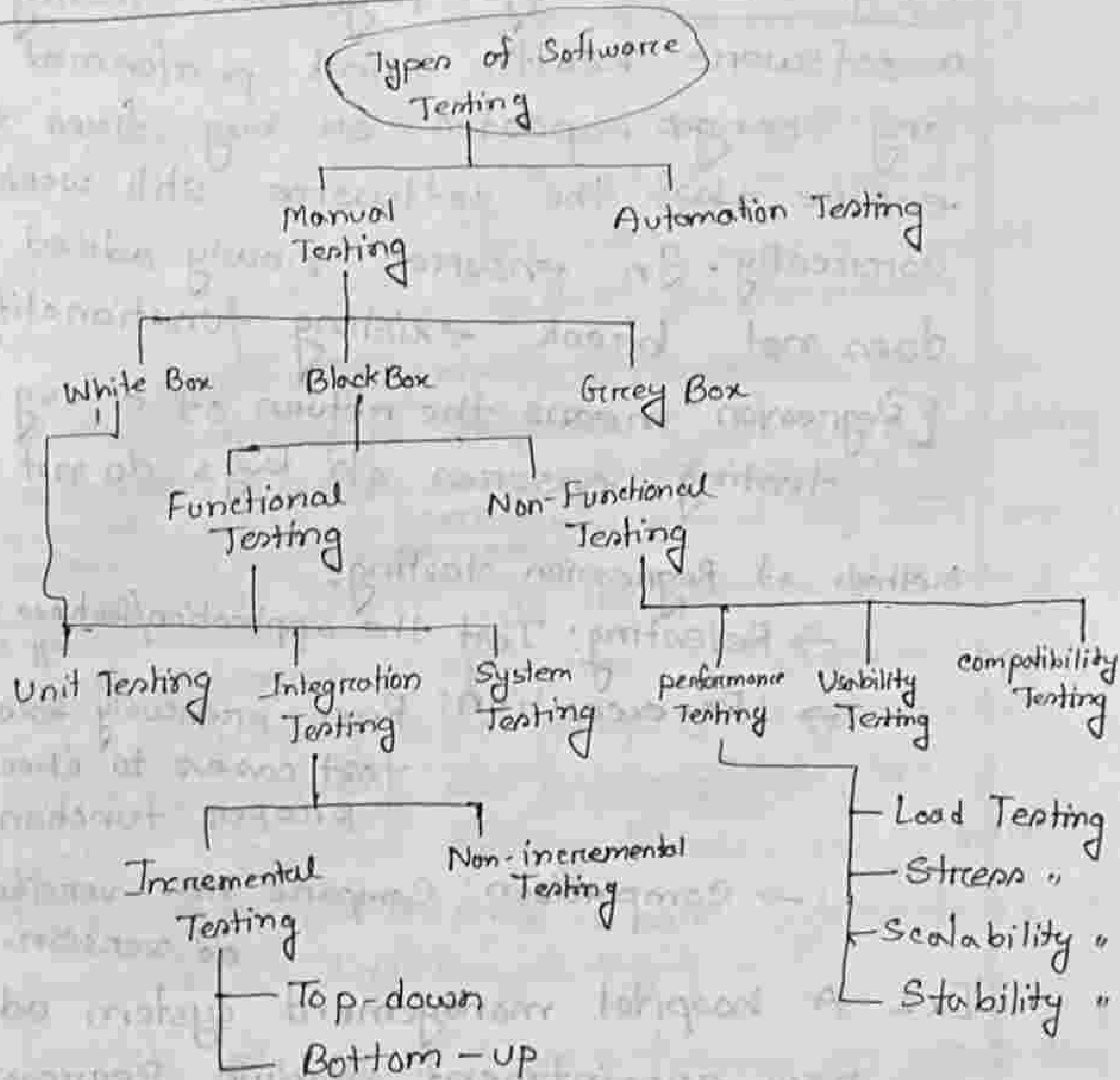
Equivalence class:

1. $x > y$ case: Loop executes with subtraction until equal.
2. $y > x$ case: Loop executes with subtraction until equal.
3. $x = y$ case: Loop bypassed, returns directly.

Test Cases:

- input : $x = 20, y = 8 \rightarrow$ Output: $GCD = 4$
- input : $x = 9, y = 27 \rightarrow$ Output: $GCD = 9$
- input : $x = 15, y = 15 \rightarrow$ Output: $GCD = 15$

From GeeksforGeeks:



→ can be considered under Black-Box / Gray-Box Testing.

* Regression Testing: Regression Testing is like a software health check performed after any changes, updates, or bug fixes to ensure ~~that~~ the software still work correctly. It ensures newly added code does not break existing functionality.

[Regression means the return of a bug - this testing ensures old bugs do not reappear]

Methods of Regression testing:

→ Retesting: Test the application/features affected by changes

→ Re-execution: Run previously executed test cases to check proper functioning.

→ Comparison: Compare new version vs old version.

Ex:- A hospital management system adds a new appointment module. Regression testing ensures login, patient records, and billing still work as before.

Advantages:

- Prevent New Bugs: Updates do not break existing functions.
- Maintains Reliability: Confirms system continues to work as expected.
- Improves stability: Helps maintain overall system performance.
- Supports Component Integration: Ensures newly added modules work with existing

Disadvantages: Time-consuming, High cost, Repetitive work.

Repetitive work, can't detect new types of bug alone, Require proper test case management.

* Integration Testing: is the process of testing

combined modules together to ensure they work correctly as a group. It is performed after unit testing and before system testing.

[When integration testing focuses on internal logic, code paths, and module interactions, it becomes a white-box testing.]

[When integration testing focuses on what the system does rather than how it does it, it is treated as functional/Black-Box Testing]

ex- In a hospital management system:

Patient registration module + Billing module

Test: When a new patient is registered, the billing

module correctly generates an invoice

Advantages:

→ Detects interface & communication errors early.

→ Helps ensure smooth interaction between modules

→ Reduces risk of system failure during later stages.

Disadvantages:

→ Time-consuming

→ Hard to locate the exact source of error.

→ Requires well defined test plans & environment

Integration testing has 4 types:

1. Top-down

2. Bottom-up

3. Sandwich

4. Big-Bang

* Performance Testing: Performance Testing is a type of software testing that checks whether ~~ensures~~ software performs well under its properly under their expected workload. It evaluates speed, responsiveness, stability, and resource usage under normal & peak conditions.

Ex- A hospital management system is ~~tested~~ tested to handle 1000 concurrent users without slowing down.

A website is tested for response time during peak traffic hours.

Advantages:

→ Ensure System Efficiency - check speed, load capacity, responsiveness

→ Identifies Bottlenecks - Detects slow database queries, memory issues, or network problems.

→ Optimizes Software - Help developers improve system

→ Ensures Enhances Reliability & Stability

→ Improves Customer Satisfaction.

Disadvantages :

→ High cost

→ Time-Consuming

→ Complex Setup

→ Limited Scope in Early Development.

→ Expertise Required

* Security Testing: It is a type of software

testing that uncovers ^{weakness} vulnerabilities in a system and ensures that its data and resources are protected from unauthorized access or threats.

Ex-

• Testing a hospital management system for SQL injection, weak passwords, or unauthorized access to patient records

• Testing a banking app to ensure transactions can't be hacked.

Advantages:

- Identifying Vulnerabilities. ↗ Weakness
- Improving System Security.
- Ensuring Compliance.
- Protects Data and Reputation.

Disadvantages:

- High Cost
- Time-Consuming
- Requires Expertise
- Not Fully Exhaustive ↗ complete

* Alpha Testing: Alpha Testing is a type of validation testing. It is a type of acceptance testing that is done before the product is ~~see~~ released to customers.

Ex - Testing a hospital management software internally within the company before release.

Advantages: Early Bug Detection, Improved Quality, Cost-Effective, Usability insights, Requirement validation.

Disadvantages:

Limited to internal testers, Feedback may be less diverse

* Beta Testing: is conducted by real users at customer sites in a real time environment before final release. It tests the software in practical conditions with actual users.

Ex - Releasing a new app to selected users, to test in real usage scenarios before public launch.

Advantages:

→ Reduce Product Failure Risk

→ Post Launch Testing

→ Improves Product Quality

→ Cost-Effective

→ Increase customers satisfaction

Disadvantages:

- Less control over testing conditions
- Feedback may be incomplete
- Bugs found late may be costly to fix.

ST24

08-04-26

9-4-26 তারিখ
note এর নাম দেবে।

09-04-20

13-04-2026

McCabe's Complexity:

* Cyclomatic complexity:

* দুই দুই way তে বের করা যায়।

* Example

→ cyclomatic complexity

* Example:

Graph থেকে, cyclomatic complexity.

* example:

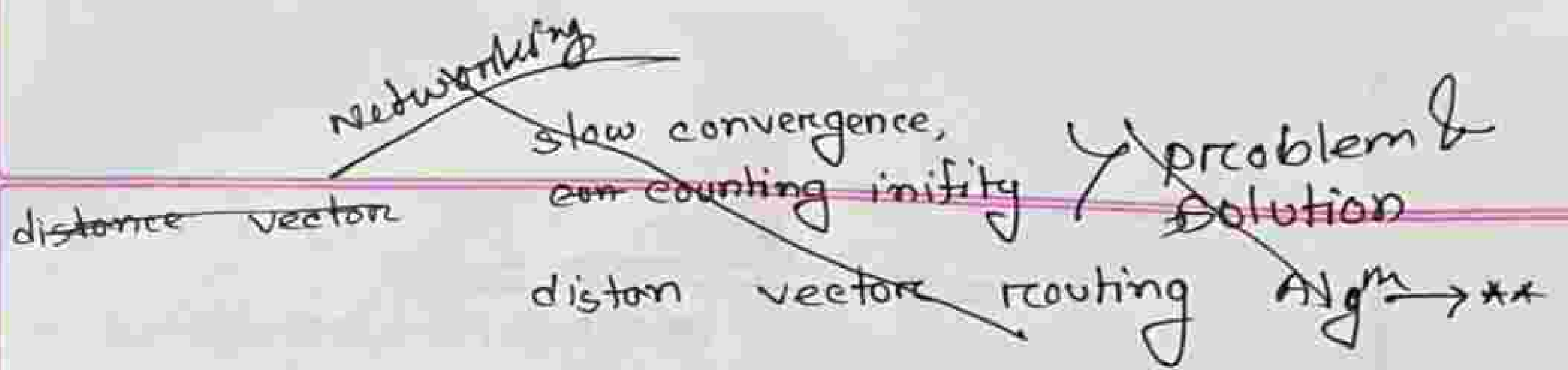
→ 367 way তে বের করতে পারি।

[~~Flow~~ code থেকে Flow chart draw
করেও বের করতে বলা হতে পারে।]

* software engineering exam Guide-এ

কিছু দিক সূচনা পাওয়া পারি-

Good software এ কি থাকে, Read কি, করা
কিছু।



08-04-2026

SE22

Measurement: Provides a mechanism for objective evaluation.

Measure: measure a number of errors.

Metric:

My Measure software?
(Slide)

- * Process Metrics: ongoing measure
Private Process Metrics → individual team
product metrics: instant measure.

* Types of measure:

* Size-Oriented Metrics:

* Halstead's Metrics:

program length, } formal
sum.

program complexity:

distinct → যতবারই আসবে তা তা count
হবে।

for exam ***

example:-

$n_1 = 8$

$n_2 = 4$

$n = 12$

এখান থেকে ২/৩ mark theory
Qstn: 4/5 marks math type
Qstn অনাগে।

SE22

09-04-2026

* Software Metrics:

* Black Box Metrics :

Function Point: Feature point & update version

~~* Size~~

* Information domain values:

* Complexity Adjustment value:

* Exercise : Function Points

→ এখানে, No influence.

↳ are not important

→ most important বসবে ৫ হবে।

parameter

value গুলো
value বাক
দাওয়া চলে গেলে
slide ৬

step 3 : value গুলো,

simple, average, complex (কিছু) (দেখা যাবে)।



→ Slide দেখানোর আর কোন থাকবে।

18-04-2026 (online class)

Risk management:

Risk mang. Paradigm:

Risk management process:
identification, analysis, planning,
monitoring, tracking & control

* The risk Management process (cont...)

* Risk Management

* Risk category

* Software risks (**)

Risk → Affects

* Reactive - Proactive Risk Mang.
↳ Risk স্বাধীন
চাপ (যে ব্যবস্থা
(নয়) হয়
→ আগ্রহ (যাক) alert থাকে

* Feature based Risk type

* Risks and risk types

* Risk Analysis

* General Risk analysis (**)

কি কারণ effect

* Risk Planning

* Strategies

* Monitoring

* Projection / Estimation

* Building a Risk Table

* Exposure

$$RE = P \times C$$

Qstn:

paradigm, 7 no. page কোন Category (৩)
affect করে (মডি. Description সহ, দুই-বিশেষ Risk
কি-কখন হয়- কিভাবে mitigate করা যায়, কোন Category
কি কারণ decision নেয়, Risk Exposure ছোট
math

[৩২ দিকে জন) Wk-1, 2, 3, 4 slide দেখা ন্যায়]